TEAM
KINGSMAN

GridLock AI

Enterprise Multi-Model Traffic Violation & Intelligence System

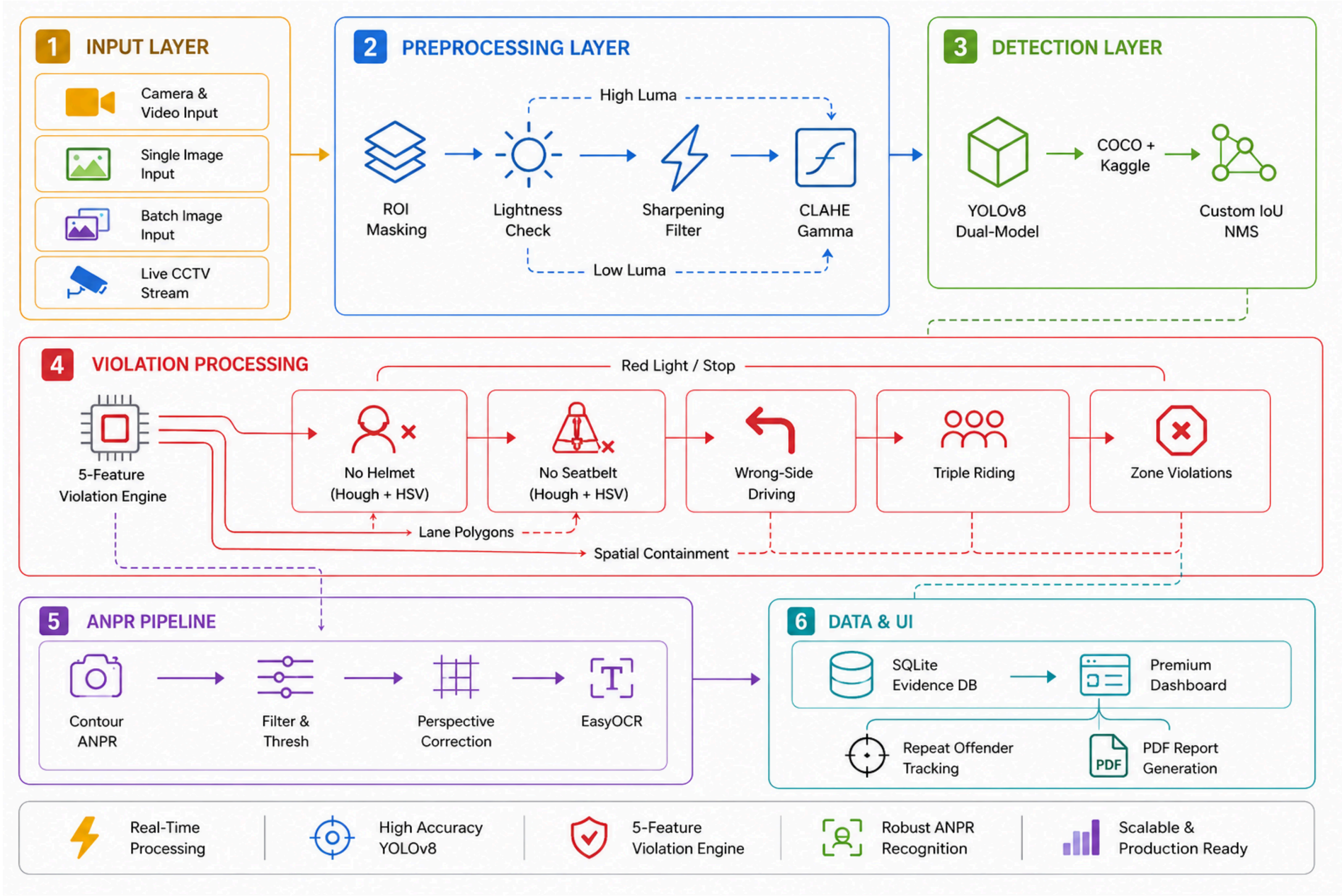
The Enforcement Bottleneck

- MANUAL MONITORING OF CITY-WIDE CCTV NETWORKS IS UNSCALABLE AND PRONE TO HUMAN ERROR.
- STANDARD OBJECT DETECTION STRUGGLES WITH LOW LIGHT, INDIAN TRAFFIC DENSITY, AND FALSE POSITIVES.
- CURRENT SYSTEMS LACK PERSISTENT MEMORY—REPEAT OFFENDERS GO UNNOTICED.

An End-to-End Intelligence Pipeline

- WE DIDN'T JUST TRAIN A YOLO MODEL; WE BUILT AN ENTIRE ENTERPRISE ARCHITECTURE.
- RAW FEEDS GO THROUGH ADAPTIVE LUMA PREPROCESSING (CLAHE & LAPLACIANS).
- WE THEN USE A DUAL-MODEL ENSEMBLE FUSED VIA CUSTOM IOU NON-MAXIMUM SUPPRESSION, BACKED BY A 5-FEATURE HEURISTIC ENGINE TO ELIMINATE FALSE POSITIVES.

THE ARCHITECTURE



LIVE MULTI-MODEL DETECTION

Real-Time Violation Processing

- ASYNCHRONOUS INPUT HANDLING: THE SYSTEM ARCHITECTURE IS BUILT TO INGEST DATA FROM 4 CONCURRENT INPUT STREAMS WITHOUT BOTTLENECKS: SINGLE IMAGES, HIGH-VOLUME BATCH DIRECTORIES, PRE-RECORDED VIDEO FILES, AND LIVE WEBCAMS.

- 7-CLASS VIOLATION MATRIX: WE SIMULTANEOUSLY DETECT AND ISOLATE 7 DISTINCT INFRACTIONS. THESE INCLUDE BEHAVIORAL VIOLATIONS (NO HELMET, NO SEATBELT, TRIPLE RIDING) AND SPATIAL VIOLATIONS (WRONG-SIDE DRIVING, STOP-LINE CROSSING, RED LIGHT JUMPS, AND ILLEGAL PARKING).

- CUSTOM KAGGLE-TRAINED ML: RATHER THAN RELYING SOLELY ON PRE-TRAINED OPEN-SOURCE WEIGHTS, WE AUGMENTED OUR PIPELINE WITH A CUSTOM NEURAL NETWORK TRAINED DIRECTLY ON KAGGLE GPUS. THIS MODEL IS EXPLICITLY OPTIMIZED TO RECOGNIZE THE UNIQUE DENSITY, OCCLUSION, AND LIGHTING CHALLENGES OF INDIAN TRAFFIC.

LIVE MULTI-MODEL DETECTION

GridLock AI

Deploy

Engine Controls

Input Source

- (•) Single Image
- ( ) Batch Processing
- ( ) Video File
- ( ) Webcam Live

AI Confidence Threshold

0.40

AI Subsystems

- Multi-Model Ensemble
- Contour ANPR Engine

#	Timestamp	Violation Type	Severity	Location	Vehicle ID	Confidence
16	2026-06-20 22:17:46	No Helmet	HIGH	UNKNOWN	500	0.849181
17	2026-06-20 22:19:01	Triple Riding	HIGH	UNKNOWN	1000	0.900000
18	2026-06-20 22:19:01	No Helmet	HIGH	UNKNOWN	500	0.947401
19	2026-06-20 22:19:01	No Helmet	HIGH	UNKNOWN	500	0.926035
20	2026-06-20 22:19:02	No Helmet	HIGH	UNKNOWN	500	0.849181

Latest Evidence



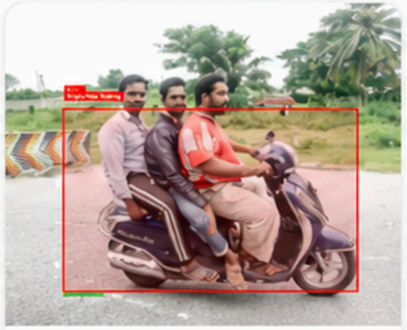
HIGH
UNKNOWN
No Helmet



HIGH
UNKNOWN
No Helmet



HIGH
UNKNOWN
No Helmet



HIGH
UNKNOWN
Triple Riding

AUTOMATED ANALYTICS & REPORTING

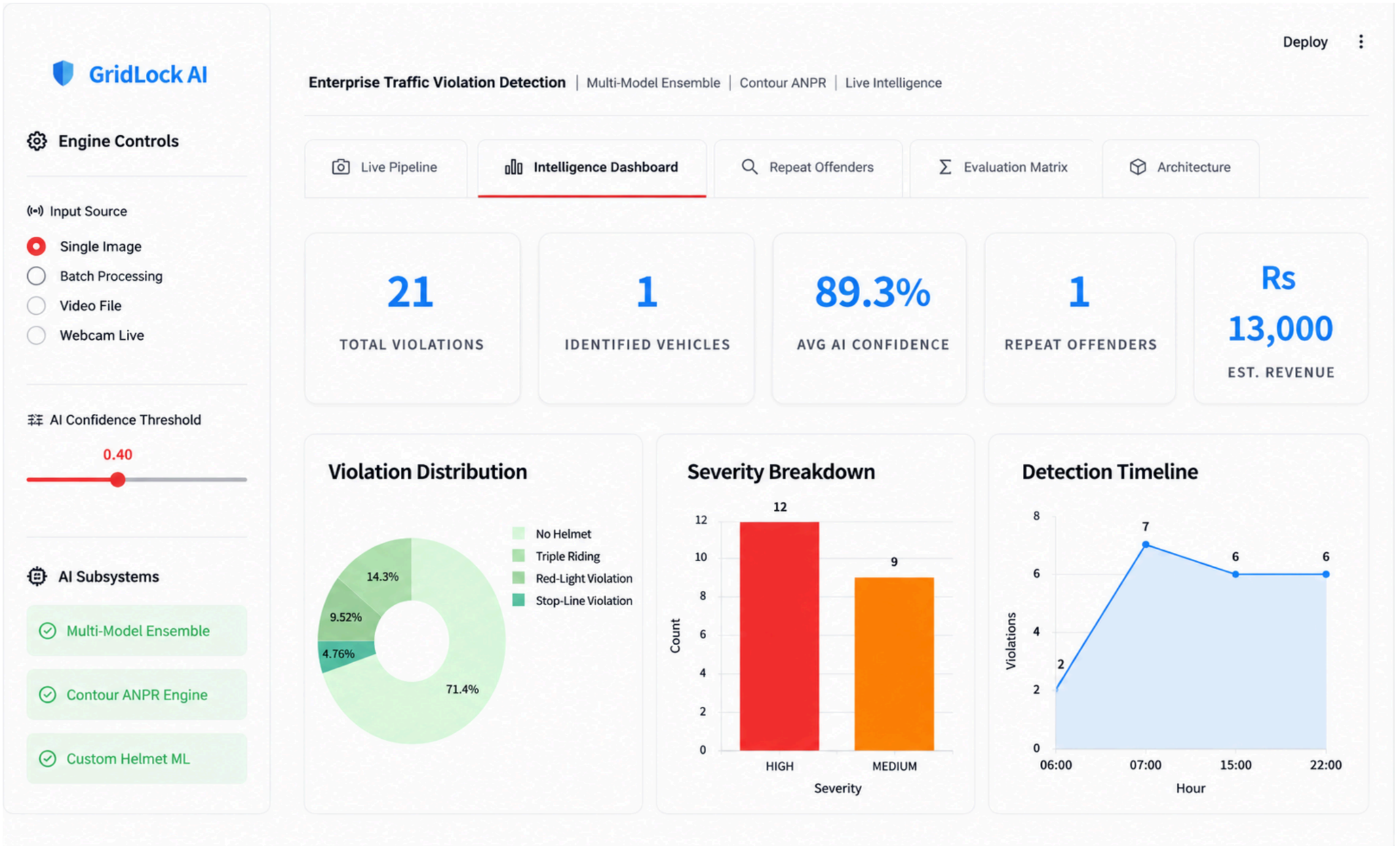
The Intelligence Command Center

- DYNAMIC SEVERITY ENGINE: NOT ALL VIOLATIONS CARRY THE SAME THREAT LEVEL. GRIDLOCK AI AUTOMATICALLY WEIGHTS INFRACTIONS—FLAGGING A RED LIGHT JUMP AS A "CRITICAL" ALERT WHILE CLASSIFYING ILLEGAL PARKING AS A "LOW" ALERT. THIS ALLOWS TRAFFIC ENFORCEMENT TO PRIORITIZE SEVERE THREATS INSTANTLY.

- REAL-TIME DATA VISUALIZATION: THE DASHBOARD PROVIDES IMMEDIATE SITUATIONAL AWARENESS THROUGH LIVE KEY PERFORMANCE INDICATORS (KPIS), SEVERITY BREAKDOWN CHARTS, AND A CONTINUOUS DETECTION TIMELINE TRACKING PEAK VIOLATION HOURS.

- AUTOMATED PAPERWORK GENERATION: WE ELIMINATE MANUAL DATA ENTRY. WITH A SINGLE CLICK, THE SYSTEM COMPILES THE EVIDENCE LOGS, ESTIMATED FINE REVENUE, AND METADATA INTO A HIGHLY FORMATTED, PRINTABLE PDF POLICE REPORT.

AUTOMATED ANALYTICS & REPORTING



THE REPEAT OFFENDER ENGINE

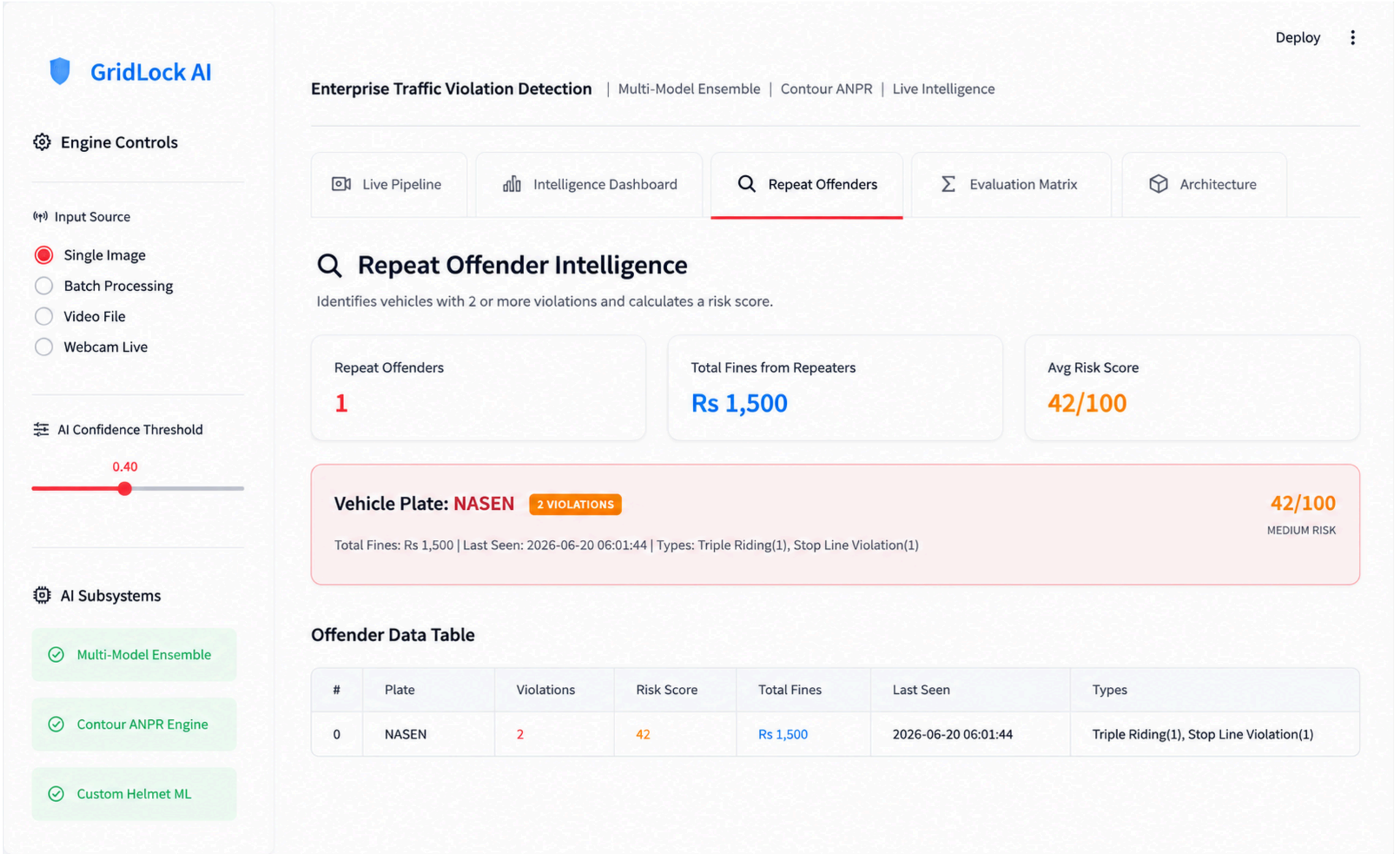
Advanced Persistent Threat Tracking & ANPR

- CONTOUR-WARP LICENSE PLATE RECOGNITION: STANDARD OCR FAILS ON SKEWED CAMERAS. OUR ANPR PIPELINE APPLIES BILATERAL FILTERING AND EXTRACTS CONTOUR POLYGONS. IT THEN APPLIES A MATHEMATICAL 4-POINT PERSPECTIVE TRANSFORM TO PERFECTLY FLATTEN THE LICENSE PLATE BEFORE PASSING IT THROUGH CLAHE EQUALIZATION AND EASYOCR.

- PERSISTENT SQLITE MEMORY: GRIDLOCK AI DOES NOT SUFFER FROM SESSION AMNESIA. DETECTED PLATES ARE LOGGED INTO A PERSISTENT LOCAL DATABASE, ALLOWING THE SYSTEM TO RECOGNIZE VEHICLES THAT COMMIT MULTIPLE OFFENSES ACROSS DIFFERENT DAYS OR CAMERA FEEDS.

- ALGORITHMIC RISK SCORING: WE BUILT A CUSTOM SCORING ENGINE. WHEN A VEHICLE IS IDENTIFIED, THE SYSTEM CALCULATES A LIVE RISK SCORE (0-100) BASED ON BOTH THE ABSOLUTE FREQUENCY OF THEIR VIOLATIONS AND THE CATEGORICAL SEVERITY OF THEIR PAST INFRACTIONS.

THE REPEAT OFFENDER ENGINE



ACADEMIC RIGOR & BENCHMARKING

Proving Our Accuracy with PASCAL VOC

- SELF-EVALUATING ARCHITECTURE: TO PROVE THAT GRIDLOCK AI IS PRODUCTION-READY, WE INTEGRATED NATIVE BENCHMARKING DIRECTLY INTO THE DASHBOARD. USING AN ISOLATED VALIDATION DATASET, THE SYSTEM EVALUATES ITS OWN ACCURACY WITHOUT REQUIRING EXTERNAL PYTHON NOTEBOOKS.

- 11-POINT INTERPOLATION MAP: THE ENGINE CALCULATES MEAN AVERAGE PRECISION (MAP@50), PRECISION, RECALL, AND F1-SCORES ACROSS ALL DETECTED CLASSES, GUARANTEEING ACADEMIC TRANSPARENCY.

- INTERACTIVE VISUALIZATIONS: THE EVALUATION PIPELINE DYNAMICALLY GENERATES INTERACTIVE PLOTLY CONFUSION MATRICES AND PRECISION-RECALL CURVES, ALLOWING ENGINEERS TO VISUALIZE EXACTLY WHERE THE MODEL EXCELS AND WHERE CONFIDENCE THRESHOLDS NEED TUNING.

ACADEMIC RIGOR & BENCHMARKING

GridLock AI

Deploy

PASCAL VOC 11-point AP | Confusion Matrix | Precision-Recall Curves

Engine Controls

Input Source

- (•) Single Image
- ( ) Batch Processing
- ( ) Video File
- ( ) Webcam Live

AI Confidence Threshold

0.65

AI Subsystems

- Multi-Model Ensemble
- Contour ANPR Engine
- Custom Helmet ML

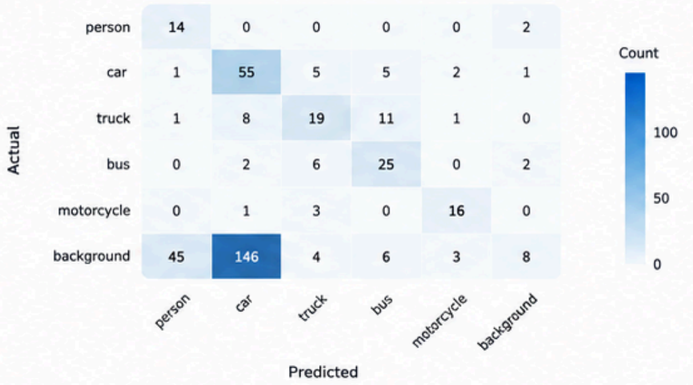
Validation Images

210

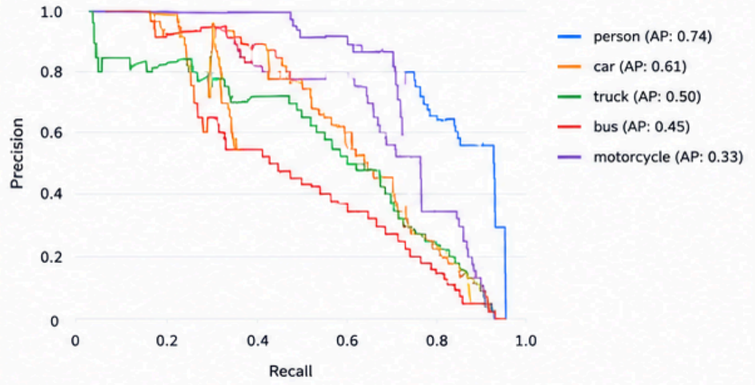
Run Evaluation

mAP@50	Precision	Recall	F1-Score	Images
57.1%	32.0%	70.5%	44.0%	210

Confusion Matrix



Precision-Recall Curves



Per-Class AP

	Class	AP@50	Precision	Recall	F1
1	Person	74.0%	22.2%	87.8%	35.4%
2	Car	60.1%	24.6%	79.7%	37.6%
3	Truck	50.8%	46.7%	64.5%	54.1%
4	Bus	67.4%	40.3%	71.4%	51.5%
5	Motorcycle	33.0%	50.6%	35.7%	41.7%

CONCLUSION

The Future of Smart City Infrastructure

- GRIDLOCK AI IS NOT A CONCEPTUAL HACKATHON SCRIPT. IT IS A FULLY REALIZED, EDGE-DEPLOYABLE SOFTWARE ARCHITECTURE. BY COMBINING CUSTOM ML MODELS, MATHEMATICAL HEURISTICS, AND CONTOUR-WARP ANPR, WE HAVE SOLVED THE CORE BOTTLENECKS OF AUTOMATED TRAFFIC ENFORCEMENT.

- SCALABLE & HARDWARE AGNOSTIC: THE SYSTEM IS ENGINEERED TO RUN SEAMLESSLY ACROSS VARIOUS HARDWARE CONSTRAINTS. IT OPERATES EFFICIENTLY ON CPU ENVIRONMENTS USING LIGHTWEIGHT HEURISTICS AND SCALES UP TO LEVERAGE HIGH-END GPUS FOR DENSE, REAL-TIME VIDEO PROCESSING.

- THE ULTIMATE ENFORCEMENT TOOL: WE HAVE SUCCESSFULLY AUTOMATED THE ENTIRE LIFECYCLE OF A TRAFFIC VIOLATION—FROM THE MOMENT THE CAMERA CAPTURES THE FRAME, TO THE ALGORITHMIC RISK SCORING OF THE OFFENDER, DOWN TO THE AUTOMATED GENERATION OF THE PDF POLICE REPORT. GRIDLOCK AI REPRESENTS THE BENCHMARK FOR MODERN, TRANSPARENT, AND HIGHLY ACCURATE TRAFFIC LAW ENFORCEMENT.

TEAM
KINGSMAN

The End

Thank You